



Question-writing: Developing Skills

Examples from Chemistry

Janet Steadman
js3064@cam.ac.uk

What are you trying to achieve?

Information for you:

Have these students understood material previously taught?

- For immediate feedback
- For subsequent lesson planning for these students
- For adjustments to future teaching

Do these students have any particular weaknesses?

- Individual or group level

What level are these students working at?

- For assessing a group new to you
- For setting
- For predicted grades
- For team selection

What are you trying to achieve?

Outcomes for students:

During the course

- Embedding techniques / knowledge
 - Building confidence in particular areas
 - Improving understanding through challenge
- Learning new material through questioning
- Feedback on their understanding / current level of attainment

Towards exams

- Improving exam outcomes through practice of specific styles
- Improving awareness of common pitfalls (stumbles)

Check the audience & format



Year group / stage?

- Any setting by ability?
- Prior knowledge?

Small group which you know well? or large (e.g. year) group?

Who is going to mark it?

- Will it need to be moderated?
- How tight does the mark scheme need to be?

For completion...

- in school?
- for homework?
- under test conditions?

Style the question to fit the purpose



For example:

I have a balloon filled with nitrogen gas at room temperature and pressure (RTP). The volume of the balloon is 14.9 dm^3 .

1 mole of any gas occupies 24.0 dm^3 at RTP.

You have been given a periodic table.

What is the mass of nitrogen in the balloon?

KS4 Embedding and building confidence



Show your working for all calculations.

1. What is the formula of a nitrogen molecule?
2. Calculate the amount in moles of nitrogen molecules in the balloon to 3 significant figures.
3. What is the relative formula mass of a nitrogen molecule?
4. Calculate the mass in grams of the nitrogen in the balloon to 2 significant figures.
You will need to use your answers to questions 2 and 3.

Quite scaffolded and could go even further by providing prompts back to specific parts of notes.

KS4 Embedding and building confidence



Show your working for all calculations.

5. Now use the same steps to calculate the mass of argon in a balloon of the same size (14.9 dm^3).

To practice the processes.

- 6.
- a) Which of the calculations for argon gave the same answer as the calculations to find the mass of nitrogen in the balloon?

To get them thinking about how gases behave.

- b) Why were these two answers the same?

To reinforce the understanding that, under the same conditions, equal volumes of gas will contain equal amounts of gas particles.

- c) Calculate the mass of bromine in another balloon of the same size.

Using bromine to check for any remaining confusion over molar mass of Br vs Br₂

KS4 end of topic diagnostic, grade 8/9



What is the mass of nitrogen in the balloon?

- A. 0.62g x calculated moles not mass, rtq
- B. 17g correct
- C. 8.7g x used R.A.M. for N not N_2
- D. 45g x incorrect calculation of amount in moles ($24/14.9$)
- E. 23g x as D and used R.A.M. for N

- No scaffolding
- All incorrect options drawn from common errors.
- Unlikely to get this as single question at GCSE (would be split) hence for higher levels.
- Feedback to students and diagnosis of misunderstanding very straightforward.

KS4 end of topic diagnostic, grade 8/9



What is the mass of nitrogen in the balloon?

A. 0.62g

B. 17g

C. 8.7g

D. 45g

E. 23g

Important to give feedback for all options (including correct), as MC can be guessed.

Alternatively, some wrong options could form the basis of a 'spot the error' type follow-up session or assignment

KS5 Checking confidence / understanding



Show your working for all calculations.

1. Calculate the mass in g, and hence the density of the nitrogen (in g dm^{-3}) in the balloon, and give both answers to 3 s.f.. *Check of prior knowledge of gas/mass/mole.*
2. The density of dry air at RTP is 1.20 g dm^{-3} , yet the balloon sinks rapidly rather than floating when released in dry air. Explain this observation. *Check of ability to identify assumptions*

Or

2. Cabin pressure in commercial airlines is usually about 80 kPa. Assuming the gas is ideal, and the temperature is constant, calculate the volume of the balloon if it was taken onboard.
Check of understanding of newer knowledge (ideal gas laws).

KS5 Checking confidence / understanding



Show your working for all calculations.

1. Calculate the mass in g, and hence the density of the nitrogen (in g dm^{-3}) in the balloon, and give both answers to 3 s.f..
2. The density of dry air at RTP is 1.20 g dm^{-3} , yet the balloon sinks rapidly rather than floating when released in dry air. Explain this observation.

The mass of nitrogen becomes part of a longer question which starts by checking understanding of prior knowledge.

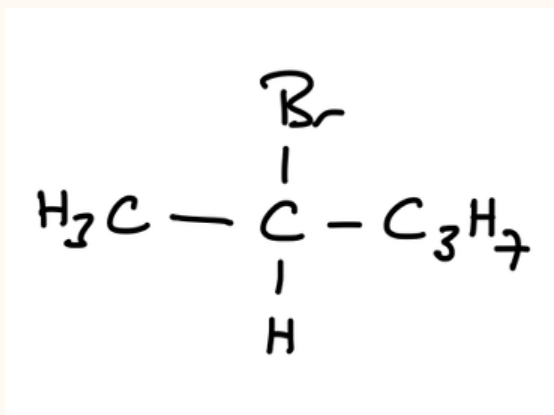
Q2 expected answer: 'mass of balloon neglected', but answer also could be pressure inside balloon $>$ RTP; can lead to discussions about assumptions made in general.

A place for 'spot the error'

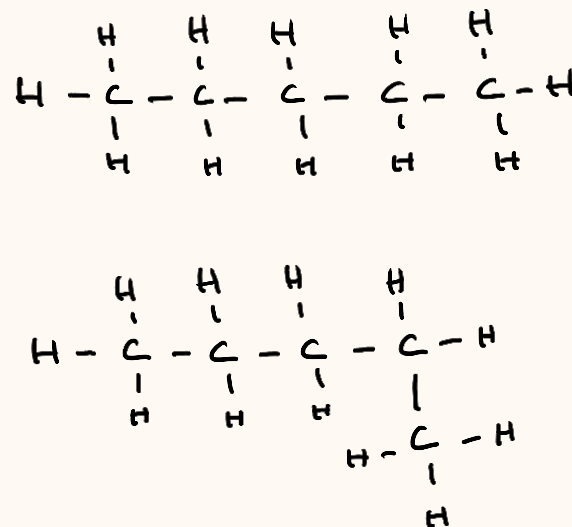


Particularly for groups which you know have particular weaknesses, or for highlighting exam points:

Draw the major product when HBr reacts with prop-1-ene



Draw two isomers of C₅H₁₂



Which is the misused word in the following explanation?

Silicon dioxide has an extremely high melting point because its giant covalent lattice structure means the intermolecular forces of attraction are very strong and require very high amounts of heat energy to break.

Random thoughts...



Writing a question where you 'know what you want them to write as an answer' is tricky, particularly for lengthy answers

- Never underestimate the ability of students to spot an alternative explanation.
- Corollary: attempts to 'ring fence' conditions can get out of hand but can be necessary (dry air vs wet air).

A really well-set diagnostic MC takes a while to write – but can definitely give back in terms of time saved

- Easy to reuse – can alter quantities for numerical questions quite quickly if necessary.
- Very quick to mark and provides rapid useful feedback to students and teachers.

Using past exam questions exclusively rarely checks full understanding.

- But has its place close to exams

Random cautions

Don't be the one who...

- Sets the question:
'Describe how to make a pure dry sample of copper(II) sulfate from copper(II) oxide powder and dilute sulfuric acid (6)'
then expects your colleagues to mark it....

Do work your questions, ideally after you've gone away and come back to them

- That way you don't set 'draw all the isomers of $C_5H_{10}O$ which contain a $-OH$ group'.

Enjoy setting questions, but beware of being drawn into random information holes

- OTOH I now know how to measure the metabolism of a goldfish...